



A Novel Natural Language Processing (NLP)–Based Machine Translation Model for English to Pakistan Sign Language Translation

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Abstract

Background/Introduction The deaf community in the world uses a gesture-based language, generally known as sign language. Every country has a different sign language; for instance, USA has American Sign Language (ASL) and UK has British Sign Language (BSL). The deaf community in Pakistan uses Pakistan Sign Language (PSL), which like other natural languages, has a vocabulary, sentence structure, and word order. Majority of the hearing community is not aware of PSL due to which there exists a huge communication gap between the two groups. Similarly, deaf persons are unable to read text written in English and Urdu. Hence, the provision of an effective translation model can support the cognitive capability of the deaf community to interpret natural language materials available on the Internet and in other useful resources.

Methods This research involves exploiting natural language processing (NLP) techniques to support the deaf community by proposing a novel machine translation model that translates English sentences into equivalent Pakistan Sign Language (PSL). Though a large number of machine translation systems have been successfully implemented for natural to natural language translations, natural to sign language machine translation is a relatively new area of research. State-of-the-art works in natural to sign language translation are mostly domain specific and suffer from low accuracy scores. Major reasons are specialised language structures for sign languages, and lack of annotated corpora to facilitate development of more generalisable machine translation systems. To this end, a grammar-based machine translation model is proposed to translate sentences written in English language into equivalent PSL sentences. To the best of our knowledge, this is a first effort to translate any natural language to PSL using core NLP techniques. The proposed approach involves a structured process to investigate the linguistic structure of PSL and formulate the grammatical structure of PSL sentences. These rules are then formalised into a context-free grammar, which, in turn, can be efficiently implemented as a parsing module for translation and validation of target PSL sentences. The whole concept is implemented as a software system, comprising the NLP pipeline and an external service to render the avatar-based video of translated words, in order to compensate the cognitive hearing deficit of deaf people.

Results and Conclusion The accuracy of the proposed translation model has been evaluated manually and automatically. Quantitative results reveal a very promising Bilingual Evaluation Understudy (BLEU) score of 0.78. Subjective evaluations demonstrate that the system can compensate for the cognitive hearing deficit of end users through the system output expressed as a readily interpretable avatar. Comparative analysis shows that our proposed system works well for simple sentences but struggles to translate compound and compound complex sentences correctly, which warrants future ongoing research.

Keywords Machine translation · Natural language processing · Deaf people communication · Pakistan Sign Language · Cognition · Rule-based translation

Introduction

Technology has always been used as a tool to facilitate people with special needs [1]. Over 5% of the world's population comprises of hearing-impaired people, which is a total of 466 million, while it is estimated that in 2050 this number will be increased to 900 million people. Deaf people use a unique language of gestures to communicate among themselves,

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which is called sign language. A gesture is a body language that comprises of hand shapes, location, movements, and some non-manual features. The non-manual features mostly include facial expressions that may include lifting eye-brows, smile, eye blinks, and mouth and tongue movements [2]. The sentence structures of sign languages are different from those of written languages; therefore, the deaf community is unable to read and understand the text written in natural languages. This challenges the cognitive abilities of the deaf community to benefit from any written material.

Global sign language does not exist and every country has its own sign language [3]. In addition, a variety of regional sign languages exist within a country. The developed countries have been working to create inclusive communities by developing new and improving existing assistive technologies for their people with disabilities, including the deaf community. A significant amount of research has been performed on the development of communication tools and strategies for American Sign Language (ASL), British Sign Language (BSL), and many other languages. However, it is important to consider that majority of the deaf population lives in the developing countries. A number of developing countries have also recently begun focusing on improving assistive technology for their deaf populations. Some excellent research work has been published that focuses on Arabic Sign Language (ArSL), Indian Sign Language (ISL), and several other sign languages. But, no significant work on standardisation and translation of Pakistan Sign Language (PSL) has been accomplished.

In order to improve the cognitive abilities of the deaf community of Pakistan, this work aims to propose a machine translation model that translates natural language text into Pakistan Sign Language. The following main challenges have been identified in this regard:

- Absence of a substantial sentence level corpus that may help in translating several types of English sentences into PSL
- Lack of availability of linguistic information for PSL
- No written grammar rules of PSL

This research chooses English as a natural language, as it is linguistically well-studied language as compared with any other scripted natural languages. It is important to mention here that Urdu, Pakistan's national language, has not been used as the source language because it does not have adequate linguistic resources in the current state-of-the-art to properly process the Urdu sentences, while another important reason is that the end product of the proposed translation system pipeline is sign language avatar which plays PSL gestures for the words in translated sentence in given sequence. Thus, it does not affect whether the intermediate translated output is in Urdu or in English language. Such translation of natural language

text into PSL gestures will be a key step towards the integration of Pakistan's deaf community into society, as the translation system may be used in developing several useful applications including the translation of study material, news articles, cell phone messages, and other useful information resources available in English into equivalent PSL.

To the best of our knowledge, this is the first machine translation model which translates English language sentences into equivalent PSL sentences. To this end, we provide the following: (i) a systematic approach to design and develop such system which involves people from the deaf community and PSL experts; (ii) a dataset has been generated with the help of deaf subject and PSL language experts; (iii) based on rigorous analysis of the dataset, a grammar has been defined to translate and verify PSL sentences; (iv) a translation model has been proposed and a software system has been developed using this translation model that takes English language sentence as input and converts it into equivalent PSL sentence using the defined grammar; (v) subjective and automatic evaluation of the proposed translation system has been carried out using appropriate evaluation measures, and by involving both deaf subjects and sign language experts.

The rest of the article has been presented in the following manner. “Related Works” provides relevant literature review. “Translation Process” discusses high-level details of the proposed translation process. The analysis of collected data has been presented in “Tense-Based Analysis”, while “Machine Translation” discusses the details of the established grammar rules for the translation system. Implementation details of a system that translates English language sentences into equivalent PSL sentences using the defined grammar have been discussed in “Machine Translation System Implementation”. The evaluation of the effectiveness and correctness of the developed translation system has been presented in “Evaluation and Results”. Lastly, conclusion and future directions have been presented in “Conclusion and Future Directions”.

Related Works

Machine translation has been an important subject in computer science for almost sixty years. During most of this time, however, machine translation for natural to sign languages has been largely overlooked. It is only within the last decade that there is a renewed interest in the automated translation for sign languages. It is linked to both progresses in linguistic studies in sign language [4, 5] and an increase in computing resources. Machine translation from natural to sign languages is pertinent for seamless integration of the deaf community into society [6]. Several such domain specific translation systems have been developed for different languages including American Sign Language [7–10], British Sign Language [11],

Spanish Sign Language [12, 13], South African Sign Language [14], Italian Sign Language [15], and German Sign Language [16].

Direct translation system has been used in TESSA system to translate English Text into BSL gestures [17], while most of the translation systems are based on rule-based or grammar-based translation; for instance, ZARDOZ [18], ASL Workbench [19], and ViSiCAST [20] are few systems for English-based sign languages. Similarly, there exists a grammar-based translation system for South African sign language [14], and for Spanish sign language [21]. Another variant of machine translation systems is statistical machine translation [22]. But, not many sign language translation systems have been developed using statistical machine translation models, as they build upon sequence of words in large data sets, which are not available for most of the sign languages. Translation systems for ASL and BSL have been designed and are getting mature. However, in recent years tools and technologies have also been developed for the translation of specific European and Asian sign languages, for instance, Greek Sign Language [23], Spanish Sign Language [13], Arabic Sign Language [24], Vietnamese Sign Language [25], Indian Sign Language [26], Bangla Sign Language [27], and Thai Sign Language [28].

Creation of Pakistan Sign Language resources and technologies is in its infancy [2]. The work begins in early 2000 [29, 30] with a study that attempted to collect data for words and phrases in PSL. Following that, however, no significant effort has been made to automate the translation of natural language into PSL. Some recent efforts have been made to layout an architectural platform to convert natural language text into PSL, and vice versa [2, 31]. But, this initiative only illustrates the key components of such a structure and does not entail any implementations.

Another orthogonal research direction is of recognising sign language gestures and converting them into text. In this regard, a significant amount of research work has been done on American Sign Language [9, 10]. Some fundamental work in this direction has also been done about PSL [32] [33]. Similarly, another relevant yet different dimension of this research is to create a standard corpus for sign language translation and gesture recognition purposes. Some basic collections for PSL gestures exist, but there is not corpus to facilitate natural to sign language translation for PSL [5].

To the best of our knowledge, the translation model proposed in this research is the first detailed and technically rich instance of machine translation from natural language to the Pakistan Sign Language. It is pertinent to mention here that rule-based translation approaches are the most suitable starting points for developing the machine translation systems in the absence of any sizeable translation corpus. Therefore, this research proposes a rule-based translation model for translating natural language into PSL. It also compiles first parallel

corpus comprising of PSL gestures, and sentences of English language and their equivalent PSL sentences. Lastly, it integrates the proposed translation model and compiled corpus into an information system that translates English sentences into PSL gestures, and evaluates the results.

Translation Process

The proposed machine translation model is based on a systematic and step by step process. Figure 1 shows high-level steps involved in the process for materialising this translation activity. The approach starts with data collection of natural language sentences, which in this case is English language. English has been selected because it is linguistically well-investigated, and presents a pool of NLP resources for various forms of natural language analysis, while data collection involves gathering a complete variety of English language sentences which are then translated into PSL by involving different deaf people, interpreters, and PSL experts. A thorough study of the collected data is carried out to determine the similarities and variations between the English and PSL language structures. It is evident from the translated data that the English and PSL sentence structure is different. The PSL grammar rules are eventually developed and formalised using context-free grammar. Finally, the effectiveness of the proposed grammar is evaluated by translating a different set of English sentences into PSL with the help of an automated translation process. The details of all these steps have been presented in the following sections and subsections.

Data Collection of Source Language

For any machine translation model to work, it is necessary to have a clear understanding of the syntactic and semantic structures of both source and target languages. The data collection process for this research has been driven by the taxonomy of English language sentences as shown in Fig. 2, which shows that the sentences are precisely categorised in terms of

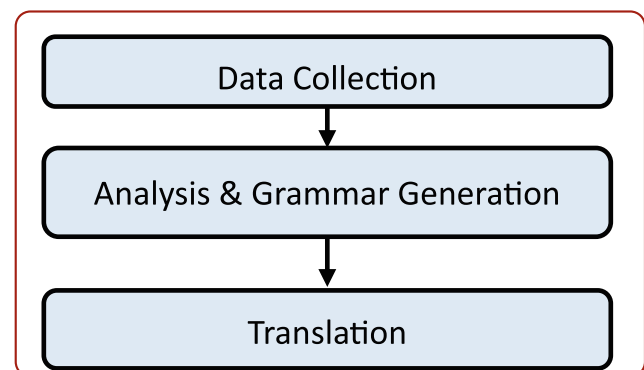


Fig. 1 Translation and evaluation

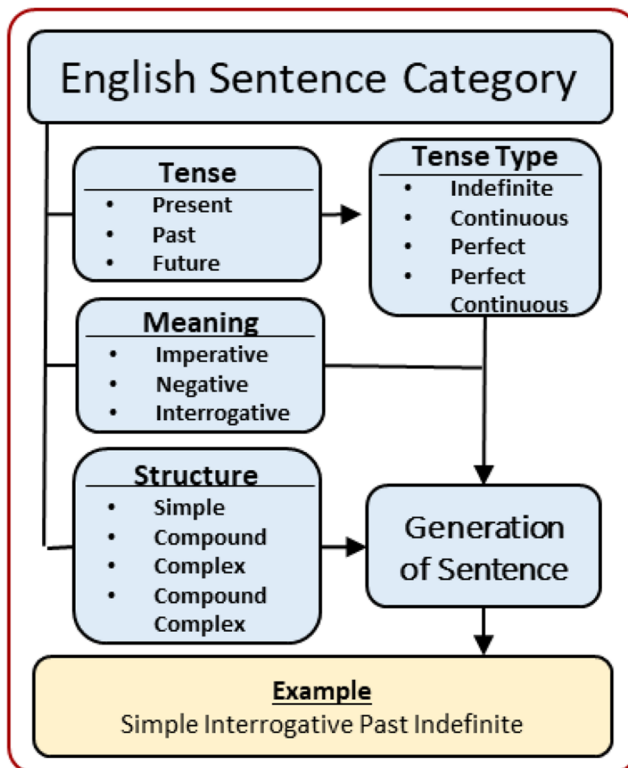


Fig. 2 English language taxonomy

structure, meaning, and tenses, whereby the tenses refer to an event's time and are divided into present, past, and future. Each tense type has four additional variations which are indefinite, continuous, perfect, and perfect continuous, while the structure refers to various degrees of complexity when composing a sentence. In terms of structure, there are four sub-categories, namely simple, compound, complex, and compound-complex. Finally, in terms of meanings they are categorised into declarative, interrogative, exclamatory, and imperative sentences.

This sentence type-based taxonomy is used to produce a rich and detailed representative dataset of English sentences. In addition to the tense, structure, and meaning variations, many widely used parts of speech, including adverbs, pronouns, adjectives, articles, negations, and multiple variants of interrogative sentences, are also included in the dataset. This leads to the creation of a source dataset consisting of about 300 English sentences covering all the various components and dimensions of the English language taxonomy shown in Fig. 2.

Translation of Dataset in PSL

This step involves a process for translating collected English language sentences into PSL sentences with the help of sign language interpreters and deaf subjects. The process is shown in Fig. 3, where different deaf subjects and interpreters have

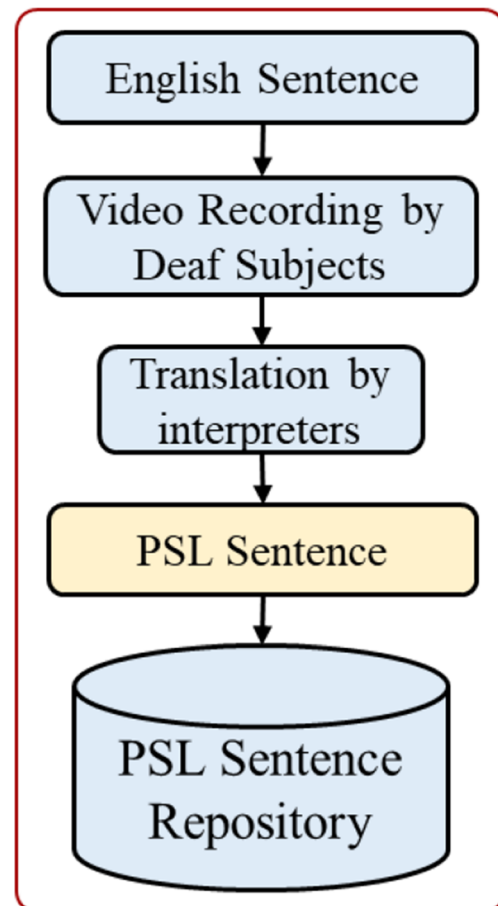


Fig. 3 PSL sentence generation

been involved in performing gestures for English language sentences into PSL.

Each source language sentence is translated into PSL by involving minimum three deaf subjects. This helps in capturing different possible variations of word order used by the deaf community in the formation of equivalent PSL sentences for provided English sentences. The gestures corresponding to sentences are recorded for future reference and analysis. Table 1 presents the translation of some sample sentences of English to their equivalent PSL versions along with their word order. To the best of our knowledge, this translated dataset is the first sentence level PSL data repository containing a wide variety of sentences.

Analysis of translated PSL dataset

The identification of the differences in English language sentences and PSL sentences requires a detailed analysis of the collected data. Therefore, the translation results are thoroughly analysed based on several linguistic heads, including word order of subject, object, and verb; tense; and sentence type based on structure, meaning, and other variations. The detailed analysis is presented in the following subsections.

Table 1 Sample translation of English to PSL

Category of sentence	English sentence and structure	PSL sentence and structure
Present indefinite	S+V+O Ali goes to school	S+O+V Ali school go
Present continuous	S + HV + V(ing) + O Ali is going to school	S + O + V + now Ali school go now.
Present perfect	S + (has/ have) + 3rd V+ O Ali has gone to school	S + O + V(Ist form) + full Ali school go full.
Future indefinite	S + Neg + V Ali will go to school	S + V + Neg Ali school go after
Past indefinite	S + adj + O This is a correct algorithm.	S + O + adj This algorithm correct.
WH-interrogative	Wh + S + V Why Ali is going?	S + V + Wh Ali going why?
Aux interrogative	Aux + S + V + O Do you want money?	S + O + V + Aux You money want Yes-No?

Sentence structure

Most of the English language sentences use subject, verb, and object (S+V+O) word order in sentence formation, whereas the data which has been obtained from the deaf subjects shows that the deaf use different subject, verb, and object permutations. An example PSL sentence, with different subject, verb, and object variations, has been presented in Table 2. The frequency of each permutation of subject, verb, and object is computed to figure out which variant is most frequently used by the deaf community. The results reveal that more than 80% of the sentences collected from the deaf community followed the (S+O+V) word order. Therefore, (S+O+V) is considered to be the default word order for PSL sentences.

Grammatical Differences Between English and PSL It is pertinent to discuss the grammatical differences between English and PSL. The data shown in Table 1 helps us in identifying the differences between the sentence structures of English and PSL in many different ways, which is discussed below:

Removal of Articles and Propositions: Written languages like English use articles and prepositions in their written scripts, while the gesture-based languages do not include such

constituents. This can be observed from the sentences shown in Table 1.

Adjectives and Adverbs: The sample sentences in Table 1 show that in PSL, the adjectives are always placed after the noun in the sentence, whereas the position of adverb remains the same as it was in the source language sentence.

Negations in PSL: In all types of PSL sentence negation is represented by adding the word “not” at the end of the sentence.

Lemmatized Form of Words: Pakistan sign language sentences are formed using only the lemmatized words. PSL does not use linking verbs and suffixes.

Time Words: Words like tomorrow, yesterday, or soon are used in English to indicate the time of an event. If any of the time words exist in the English sentence, it must be placed at the beginning of the sentence in PSL.

Interrogative: Sentences having questions in them are called interrogative sentences. They are further divided into two subcategories:

1. *Wh* question, if the source sentence is an interrogative then it's *wh*-question part is moved to the end of the sentence, except if it starts with “when”, and in that case, the word “time” is placed at the end of the sentence.
2. Auxiliary question: if the source sentence is an interrogative, then its auxiliary question part is replaced by “yes no” and is moved to the end of the sentence.

Part of Speech (POS) Tagging

Penn Treebank contains part of speech tags, and there are about 48 different POS tags available in Penn treebank. The analysis of translated PSL sentences reveals that in PSL sentence formation only employs basic forms of POS tags unlike richer POS tags used for English sentences. Figure 4 shows

Table 2 PSL word order permutations

Sentence	Word order
Like badminton she.	V+O+S
She badminton like.	S+O+V
She badminton like.	S+O+V
Badminton like she.	O+V+S
Badminton she like.	O+S+V
Like she badminton.	V+S+O

Fig. 4 POS Tags for English and corresponding PSL sentence

	My	Cat	is	Chasing	a	mouse
ENGLISH	PRP	NN	VBZ is-	VBG	DT a	NN
		cat- nsubj	aux	chasing	det	mouse dobj
PSL	My	Cat	Mouse	Chase	Now	
	PRP	Noun nsubj	NN obj	Verb chase root		

POS tags for an example English sentence along with simple POS tags of mapped PSL sentence. It can be observed from the figure that “is” that is tagged as VBZ has been eliminated; similarly, “chasing” is assigned VBG tag which has been simplified as verb’s root form in corresponding PSL sentence. Table 3 shows the mapping of Penn Treebank POS tags for English onto POS tags used in PSL.

Tense-Based Analysis

Tenses refer to the time of an event. In the English language, tenses are primarily categorised into present, past, and future. All these three primary categories further include variants namely indefinite, continuous, perfect, and perfect continuous as shown in Fig. 2. The two other basic types of tenses are simple present and simple past, which makes it a total of 14 different categories of English sentences w.r.t. tenses. The dataset contains at least ten sentences of each mentioned category of tenses along with their PSL translations which has

been collected from deaf subjects and validated by sign language experts and interpreters. This section provides a detailed overview of these various types of tenses while highlighting differences for each tense between English and PSL sentences.

Simple Present and Simple Past Tense

This type of tense is used to represent basic facts, e.g. “I am tired” and “He was sick”. Table 4 shows the translation of simple present and simple past tense sentences covering positive, negative, and interrogative variants.

Following interesting observations are evident from Table 4. S+O+V sentence structure has been observed both in present and past tense sentences of PSL. It can also be observed that helping verbs, articles, and propositions have been removed during the translation into PSL sentences for both tenses. Another important observation is that in case of simple past tense, the word “was” has been added at the beginning of the translated sentence to represent past tense. Similarly, the negation in PSL is shown by adding the word “not” at the end of the sentence for both tenses, while for the auxiliary-interrogative sentences “yes-no” is added at the end of the translated sentence.

Present, Past, and Future Indefinite Tense

The analysis for the present, past, and future indefinite tenses reveals the following observations. In English, the present indefinite affirmative sentences add a suffix “s” or “es” to the base form of the verb when the subject is singular, while for a plural subject only base form of the verb is used as shown in the examples: “Ali plays a piano”; “They sing the national song”. Similarly, for present indefinite negative sentences “does not” or “do not” is used along with the base form of the verb, e.g. “Ali does not play a piano” and “They do not sing the national song”, while in the case of present indefinite interrogative sentences either “does” or “do” comes at the beginning of the sentence along with the base form of the verb, e.g. “Does Ali play a piano?” and “Do they sing the national song?”.

Table 3 PSL tags corresponding to pen tags

Penn tags	PSL tags
NN	NOUN
NNS	
NNP	
NNPS	
PRP	VERB / AUX
VB	
VBD	
VBG	
VCN	
VBP	
VBZ	
JJ	ADJ
JJS	
NP	SUBJ/OBJ/NP
IN	
TO	PREP
RP	

Table 4 Sample PSL translation of simple present and past tense

Category of sentence	English sentence	PSL sentence
Simple present affirmative	He is a boy.	He boy.
Simple present negative	He is not clever.	He clever not.
Simple present interrogative	Is he clever?	He clever Yes-No?
Simple past affirmative	He was the man of words.	Was he words man?
Simple past negative	He was not the man of words.	Was he words man not?
Simple past	Was he the man of words?	Was he words man Yes-No?

The translated sentences show that regardless of a singular or plural subject, PSL uses only the root form of the verb. For articles, prepositions, negation, and interrogation, the same rules hold which have been drawn in simple present and past tense cases earlier; also the words “does” or “do” also get eliminated in the translated PSL sentences.

In English the past indefinite affirmative sentences consist of simple-past form of the verb as shown in the examples: “Ali played a piano”; “They sang the national song”. While for past indefinite negative sentences, “did not” is used along with the base form of verb, e.g. “Ali did not play a piano” and “They did not sing the national song”. Similarly, in the case of past indefinite interrogative sentences, “did” comes at the beginning of the sentence along with a base form of the verb, e.g. “Did Ali play a piano?” and “Did they sing the national song?”.

The translation of these different variants of past indefinite sentences into PSL reveals one commonality; i.e. the word “was” has been added at the start of the sentences, while the simple-past form of the verb changes to its base form, whereas no change has been observed for negative and interrogative sentences. Lastly, the word “did” also gets eliminated from the translated sentences. The translation for different variants of the present, past, and future indefinite sentences has been presented in Table 5. It can be observed that the word “after” is added at the end of each sentence to reflect that the task performed by the subject may complete later w.r.t time. Furthermore, the words “will” and “shall” which are used to

represent future tense in English sentences have been eliminated in the translated PSL sentences. It is quite interesting to note that in PSL, the sentences belonging to past and future tenses follow the same grammatical structure that has been observed in the present tense, while additional words like “was” and “after” are added for past and future tenses, respectively.

Present, Past, and Future Continuous Tense

The present continuous tense formation rules in English involve the usage of helping verbs like “is”, “am”, “be”, and “are” along with applying the suffix “ing” to the verb. This is also known as present participle form of the verb. The data of different variants of continuous tense has been presented in Table 6, and it shows that for continuous tense the word “now” is included at the end of the PSL sentences, while helping verbs along with articles and propositions are omitted from the translated PSL sentence. Likewise, the present participle form of the verb is converted back to the simple-present form of the verb. Lastly, the words “was” and “after” are used for past and future tenses, respectively.

Present, Past, and Future Perfect Tense

Perfect tense sentences are considered the next category. In English language the sentences belonging to this group use “has” or “have” for the present, “had” for past, and “will have”

Table 5 PSL Translation of simple present, past, and future indefinite tense

Category of sentence	English sentence	PSL sentence
Present indefinite affirmative	He goes to school.	He school go.
Present indefinite negative	He does not go to school.	He school go not.
Present indefinite interrogative	Does he go to school?	He school go Yes-No?
Past indefinite affirmative	He went to school.	Was he school go.
Past indefinite negative	He did not go to school.	Was he school go not.
Past indefinite interrogative	Did he go to school?	Was he school go Yes-No?
Future indefinite affirmative	He will go to school.	He school go after.
Future indefinite negative	He will not go to school.	He school go after not.
Future indefinite interrogative	Will he go to school?	He school go after Yes-No?

Table 6 PSL translation of simple present, past, and future continuous tense

Category of sentence	English sentence	PSL sentence
Present continuous affirmative	He is going to school.	He school go now.
Present continuous negative	He is not going to school.	He school go now not.
Present continuous interrogative	Is he going to school?	He school go now Yes-No?
Past continuous affirmative	He was going to school.	Was he school go now.
Past continuous negative	He was not going to school.	Was he school go now not.
Past continuous interrogative	Was he going to school?	Was he school go now Yes-No?
Future continuous affirmative	He will be going to school.	He school go now after.
Future continuous negative	He will not be going to school.	He school go after not.
Future continuous interrogative	Will he be going to school?	He school go after Yes-No?

or “shall have” for future tenses, whereas these sentences use past-participle forms of the verbs. Some sample translated sentences for different variants of perfect tense have been presented in Table 7. It can be observed from the translated sentences that for all tenses the word “full” has been added at the end of the sentence in PSL, whereas the use of “*was*” and “*after*” for past and futures tenses remains consistent as previous observations.

Present, Past, and Future Perfect Continuous Tense

In perfect continuous tenses, the phrases “has been”, “have been”, “had been”, “will have been”, or “shall have been” are used along present form of the verb. The examples of translation of perfect continuous tense have been presented in Table 8. It can be observed that even in these sentences, the PSL sentences use the basic S+O+V structure, while the translated sentences follow the same rule that have been observed in the continuous and the perfect tense categories, i.e. the addition of “full” and “now” in the sentences.

Machine Translation

Machine translation (MT) is also known as computerised translation. During the machine translation process, automated software techniques are used to translate the text written in one natural language such as English to another, such as Urdu.

Machine translation of natural language to sign language is relatively a new area of research which is gaining importance in recent years. For any kind of translation, the essential requirement is to preserve the meaning of source language sentences after converting it into the target language. Machine translation is mainly classified into semantic, statistical, and neural translation models [8, 33].

Statistical and Neural Machine Translation Models These techniques need a bilingual dictionary consisting of sentences and their translations. Specific statistical methods manipulating the sequence of words are applied to the bilingual corpus during sentence translation [34, 35]. These models generate quick and accurate translations, but at the same time, the translation accuracy is heavily depended on the availability of such a significant corpus that covers all the grammatical categories of source and target language. These approaches cannot be applied to our system because, to the best of our knowledge, no such corpus is available for PSL. As a matter of fact, no such corpora exist for almost any other sign language as well [36].

Semantic Translation Models These models are based on rules or grammars to translate from one language to another. Following are different types of semantic translation models.

Direct Translation Models Word to word substitution is applied in this kind of translation; no grammatical and semantic

Table 7 PSL translation of simple present, past, and future perfect tense

Category of sentence	English sentence	PSL sentence
Present perfect negative	He has not gone to school.	He school go full not.
Present perfect interrogative	Has he gone to school?	He school go full Yes-No.?
Past perfect affirmative	He had gone to school.	Was he school go full.
Past perfect negative	He had not gone to school.	Was he school go full not.
Past perfect interrogative	Had he gone to school?	Was he school go full Yes-No?
Future perfect affirmative	He will have gone to school.	He school go full after.

Table 8 PSL translation of simple present past and future perfect continuous tense

Category of sentence	English sentence	PSL sentence
Present perfect continuous affirmative	He has been going to school.	He school go full now.
Present perfect continuous negative	He has not been going to school.	He school go full now not.
Present perfect continuous interrogative	Has he been going to school?	He school go full now Yes-No.?
Past perfect continuous affirmative	He had been going to school.	Was he school go full now?
Past perfect continuous negative	He had not been going to school.	Was he school go full now not?
Past perfect continuous interrogative	Had he been going to school?	Was he school go full now Yes-No.?
Future perfect continuous affirmative	He will have been going to school.	He school go full after now.
Future perfect continuous negative	He will have not been going to school.	He school go full after now not.
Future perfect continuous interrogative	Will he have been going to school?	He school go full after now Yes-No.?

understanding of target language is considered. It is quite a visible form of this fact that such type of translations only consider the source language. It has been learned from the literature survey that sign languages are natural languages with their sentence structure, grammar, and semantic rules. Therefore, this direct translation approach of word by word substitution shall not be able to preserve the semantics of the source sentence being translated.

Rule-Based Translation Models A rule-based translation system requires a collection of rules, and a software system to process those rules. An important requirement in a rule-based machine translation model is the existence of grammar for both languages so that all specific sentence categories of the source language can be translated into the target language keeping the syntax intact. Different sign languages like ASL, BSL, and ArSL are well studied, and their grammatical structures are well defined. Due to this reason, there exist a few rule-based MT systems for these languages [37, 38]. However, PSL is not linguistically well investigated and no rules are available in the literature for the formation of PSL sentences. The main challenge for English to PSL machine translation using a rule-based approach is the generation of PSL grammar. An in-depth analysis has been performed to determine the differences between the source and target language as discussed in the previous section. Furthermore, the transformation rules for translating English sentences into PSL have been identified.

It is pertinent to highlight again that English is chosen as a source natural language in this research mainly because not only the grammar of English is well documented but various computational tools are also available for the pre-processing of the source language sentences, including POS tagging and dependency illustrations of these POS. Furthermore, to translate English text to equivalent PSL sentence, there is a strong need for PSL grammar which can be used to transform an English language sentence into equivalent PSL sentence.

Need for Grammar and Its Formalism Grammar comprises of building blocks of any language's sentence structure. Every spoken language has a grammatical structure for its sentence formation. The differences and similarities between English and PSL sentences have been identified during the analysis of the sentence level corpus. Based on this analysis, generalised rules have been formulated which are helpful in proposing an appropriate grammar-based translation model. To this end, instead of implementing the rules using basic *if-else* statements, a much better idea is to use any formal notation to express these rules that can be quickly processed using some algorithmic process. In this research work, context-free grammar (CFG) has been used as a formal notation to express these translation rules. The main advantages of using grammar formalism are as follows: (a) it is extendable, and new rules can easily be added as and when required; (b) rules can be modified and fixed if they required amendments; (c) state-of-the-art natural language parsing techniques can be applied easily.

Grammar Generations

Grammar is a structure of a language, also referred to as rules of a language. Better familiarity with the grammar of a language results in clear and understandable communication in a given language. Thus, the understanding of grammar increases the effectiveness of communication. Context-free grammar has been used to represent the rules for PSL.

A context-free grammar (CFG) is a notation for describing languages. It is a set of recursive rewriting rules or productions used to generate a string of a language. Formally, a CFG consists of four tuple information $G = \{V, \Sigma, R, S\}$ and each component of the tuple is defined as follows:

1. V is a finite set; each element $v \in V$ is called a non-terminal or variable. Each variable represents a different type of phrase or clause in the sentence.

2. Σ is a finite set of terminals, disjoint from V , which makes up the actual contents of the sentence. The set of terminals is the alphabet of the language defined by the grammar G .
3. R is a finite relation from V to $(V \cup \Sigma)^*$ where the asterisk represents the Kleene Star operation. The members of R are called the (rewrite) rules or productions of the grammar.
4. S is the start variable (or start symbol), used to represent the whole sentence (or program). It must be an element of V .

An excerpt from the CFG productions which are applied on an English sentence to produce an equivalent PSL tree has been presented in Table 9. The part of CFG presented in Table 9 only handles affirmative sentences in the present, past, and future indefinite tenses of English language, along with different productions of subject, noun, verb, and adjectives.

Machine Translation System Implementation

This section presents the implementation details of English to PSL translation system using a grammar generated in the previous section. The system comprises of two main components: a pre-processing component and a translator, whereby the preprocessing component processes the input English language sentence by performing POS tagging and dependency analysis, and finally identifies the sentence type, whereas the second component uses the grammar proposed in the previous section to convert the input sentence into equivalent PSL sentence. This component involves the renaming

of POS tags, PSL tree formation using appropriate grammar productions, and PSL sentence generation, as illustrated in Fig. 4.

English sentence is an input to the translation module, and an equivalent sentence in PSL is the output. The whole process is shown in Fig. 5. In the first stage, each word in the sentence is tagged into POS using Stanford parser and Penn treebank [39, 40]. The output is in string format and is converted into a parse tree using the tree generation algorithm. In the next step, a dependency tool is used to identify the relationship between different words that constitute the sentence [41]. This dependency information is added in the parse tree and results in the construction of the annotated parse tree. As mentioned in the analysis section, deaf people have very little knowledge of English grammar. Furthermore, it is extracted from the dataset that they use very few and fundamental POS tags that are used in the grammar generated for PSL. Therefore, as the next step mapping of Penn treebank tags into PSL tags is defined using the mappings shown in Table 3, where the annotated parse tree nodes are renamed using PSL POS tags. The next step in the translation pipeline is to classify the sentence based on tense and meaning so that the appropriate rule from PSL grammar can be selected to translate the given source sentence. In order to generate PSL tree, the English parse tree is converted into its equivalent PSL tree after applying the CFG production.

During the transformation of English parse tree into PSL tree, specific tree nodes of the English language are deleted, some new nodes are added, and due to the differences in the word order of both the language the position of some nodes is

Table 9 Excerpt from grammar productions of PSL

Sentence type	Pakistan Sign Language grammar productions
Simple present affirmative	$\langle \text{SIMPLE-PRESENT-AFFIRMATIVE} \rangle ::= \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{PRPS} \rangle \langle \text{NP1} \rangle \langle \text{NP2} \rangle \langle \text{VP} \rangle \langle \text{ADVP} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle$
Simple past affirmative	$\langle \text{SIMPLE-PAST-AFFIRMATIVE} \rangle ::= \langle \text{"was"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{"was"} \rangle \langle \text{ADVP} \rangle \langle \text{">"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle$
Present indefinite affirmative	$\langle \text{PRESENT-INDEFINITE-AFFIRMATIVE} \rangle ::= \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{NP} \rangle \langle \text{ADVP} \rangle \langle \text{VP} \rangle$
Past indefinite affirmative	$\langle \text{PAST-INDEFINITE-AFFIRMATIVE} \rangle ::= \langle \text{"was"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{"was"} \rangle \langle \text{PP} \rangle \langle \text{">"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{"was"} \rangle \langle \text{NP} \rangle \langle \text{">"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{"was"} \rangle \langle \text{ADVP} \rangle \langle \text{">"} \rangle \langle \text{SUBJ} \rangle \langle \text{VP} \rangle$
Future indefinite affirmative	$\langle \text{FUTURE-INDEFINITE-AFFIRMATIVE} \rangle ::= \langle \text{SUBJ} \rangle \langle \text{VP} \rangle \langle \text{"after"} \rangle$
Subject productions	$\langle \text{SUBJ} \rangle ::= \langle \text{NOUN} \rangle \langle \text{ADJ} \rangle \langle \text{NOUN} \rangle \langle \text{PP} \rangle \langle \text{NP} \rangle \langle \text{NOUN1} \rangle \langle \text{NOUN2} \rangle \langle \text{NP1} \rangle \langle \text{NP2} \rangle \langle \text{ADJP} \rangle \langle \text{NOUN1} \rangle \langle \text{NOUN2} \rangle \langle \text{NP} \rangle \langle \text{NOUN} \rangle \langle \text{CD} \rangle \langle \text{ADJ} \rangle \langle \text{NOUN} \rangle \langle \text{PRPS} \rangle \langle \text{NOUN1} \rangle \langle \text{NOUN2} \rangle \langle \text{ADJP} \rangle \langle \text{NOUN} \rangle$
Noun phrase	$\langle \text{NP} \rangle ::= \langle \text{NOUN} \rangle \langle \text{ADJ} \rangle \langle \text{NOUN} \rangle \langle \text{PP} \rangle \langle \text{NP} \rangle \langle \text{NOUN1} \rangle \langle \text{NOUN2} \rangle \langle \text{NP1} \rangle \langle \text{NP2} \rangle \langle \text{NOUN2} \rangle \langle \text{ADJP} \rangle \langle \text{NOUN} \rangle$
Verb phrase	$\langle \text{VP} \rangle ::= \langle \text{ADVP} \rangle \langle \text{VERB} \rangle \langle \text{OBJ} \rangle \langle \text{VERB} \rangle \langle \text{PP} \rangle \langle \text{VP} \rangle \langle \text{VERB} \rangle \langle \text{PP} \rangle \langle \text{VERB} \rangle \langle \text{ADJP} \rangle \langle \text{OBJ} \rangle \langle \text{ADVP} \rangle \langle \text{VERB} \rangle \langle \text{ADJP} \rangle \langle \text{VERB} \rangle \langle \text{ADVP} \rangle \langle \text{VP} \rangle \langle \text{PP} \rangle \langle \text{VERB} \rangle \langle \text{S} \rangle$
Adjective	$\langle \text{ADJP} \rangle ::= \langle \text{ADJ} \rangle \langle \text{SBAR} \rangle \langle \text{FW} \rangle \langle \text{PP} \rangle \langle \text{ADV} \rangle \langle \text{ADJ} \rangle \langle \text{PP} \rangle \langle \text{ADV} \rangle \langle \text{VERB} \rangle \langle \text{ADJ1} \rangle \langle \text{ADJ2} \rangle$

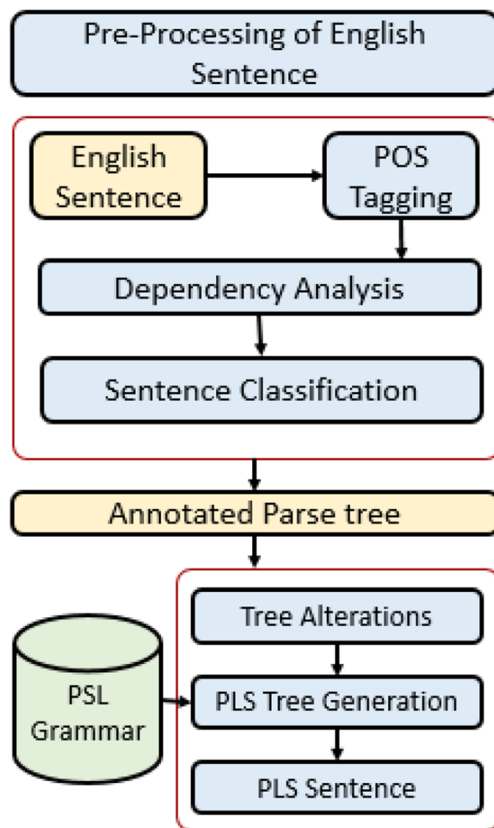


Fig. 5 Steps in translation process

altered in the destination tree. The details of all these steps have been presented in the following sections and subsections.

POS Tagging

In order to identify the linguistic properties of each individual textual unit, tagging is applied to English sentences. In this work, Stanford POS Tagger is used for tagging purpose [39]. The English sentences are given as input to the POS tagger, and along with tokenising the sentence, the tagger categorises individual tokens based on POS information such as articles, adverbs, verbs, noun, pronouns, and adjective. The tagger uses the Penn treebank tag set for tagging purpose [40]. For further processing, the string output of the tagger is transformed into a tree-like structure using the proposed string to tree conversion algorithm shown in Fig. 6. An example sentence, its POS tagging, and tree construction using the proposed algorithm have been illustrated in Fig. 7.

Dependency Analysis

The next step in the translation is to add dependency information into the tree constructed in the previous step. Stanford's typed dependencies tool gives information about the dependencies between different parts of speech involved in the construction of the sentence [40]. After the identification of

Function (NLP output string)

// to convert NLP output string into Tree Data Structure

```

1. Inputs:
2. NLP output string: A sentence of the Natural Output Language.
3. Begin.
4. String ← str NLP output string.
5. Character ← ch: A character type pointer.
6. Tree ← tree: A tree class type pointer.
7. for ← ch start of NLP output string to end of NLP output string.
8.   if ( ch equals opening brace character ) then
9.     Open_Brace_char()
10.    (Function Call for processing open Brace Characters)
11.   else if ( ch equals closing brace character ) then
12.     /* move tree pointer to current tree's parent node. */
13.   end if (and) end while
14. End.

```

Fig. 6 NLP string to tree generation

various dependencies, the textual output is transformed into a dependency graph, and later on this information is added in the parse tree which is generated in the previous step. This parse tree with dependency information is known as the annotated parse tree. The whole process of adding dependencies and generation of the annotated parse tree has been presented in Fig. 8.

Sentence Classification

It is essential to choose the appropriate PSL grammar rule to translate the English sentence to PSL sentence correctly. Before selecting a proper grammar rule, the type of English language sentence needs to be identified. As analysis revealed that PSL

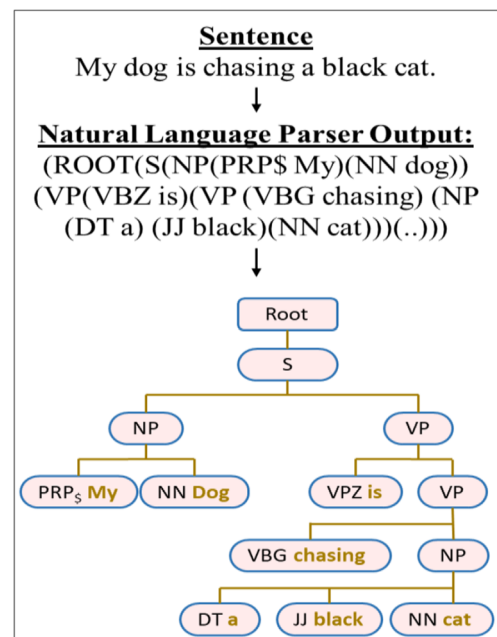


Fig. 7 POS tagging and tree generation

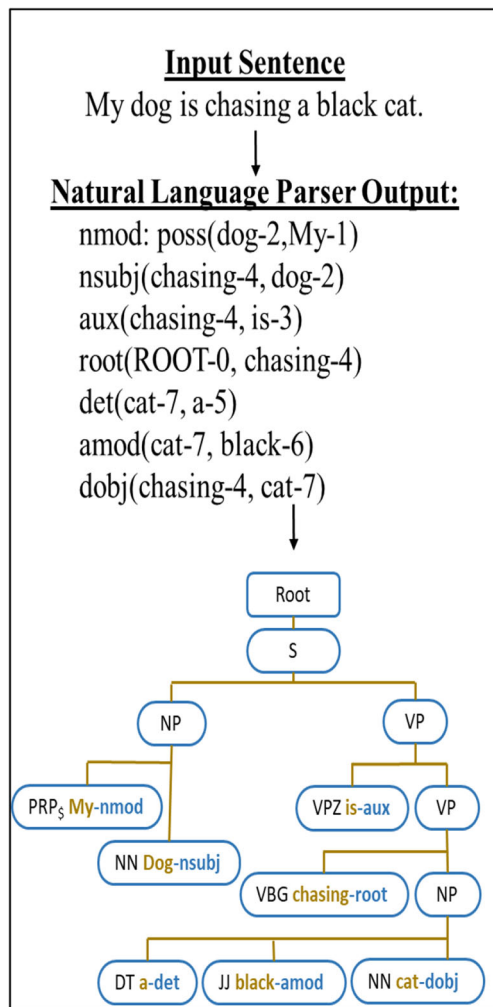


Fig. 8 Adding dependencies and generation of annotated parse tree

only uses simple sentence structure because deaf subjects lack grammatical knowledge, and due to limited vocabulary set, it is difficult for them to remember and understand complex sentence structures. As the sentences are classified based on tenses, and there are 14 different categories w.r.t. tenses available in English. So, in order to identify a specific tense type, the system determines POS tags assigned to verbs using Penn Treebank. In the example sentence, the verb part contains two tags “VBZ” and “VVG”. The first tag VBZ indicates present tense, whereas the second tag VVG implies the tense form to be continuous. By combining these two tag categories, the tense is recognised to be present continuous. Tense identification algorithm has been presented in Fig. 9.

Renaming POS Tagging It has already been discussed in the analysis section that Penn POS tagger uses a richer set of tags to annotate English sentences, whereas the corresponding PSL sentences are tagged using a simpler tag subset. Therefore, the POS tags applied by Stanford parser are replaced by relevant simplified tags for PSL.

```

1. Function (input Sentence).
2. Input
3. String ← input Sentence.
4. Start
5. String ← sentence: input Sentence.
6. if (sentence contains "shall" or "will" ) then
7.   return contains_Will/Will()
8. (Function call to process will/shall containing Sentences.)
9.   return Future Indefinite
10. else if ( sentence contains "had" ) then
11.   return contains_Had()
12. (Function call to process had containing Sentences.)
13.   return Simple Past
14. else if ( sentence contains "was" or "were" ) then
15.   return contains_Was/Were()
16. (Function call to process was/were containing Sentences.)
17.   return Simple Past
18. else if ( sentence contains "is" or "am" or "are" ) then
19.   return contains_Is/Are/Am()
20. (Function call to process is/are/am containing Sentences.)
21.   return Simple Present
22. else if ( sentence contains "has" or "have" ) then
23.   return contains_Has/Have()
24. (Function call to process had containing Sentences.)
25.   return ← Simple Present
26. else if (sentence contains "did" or 2nd form of verb) then
27.   return ← Past Indefinite
28. else return ← Present Indefinite
29. end if
30. END
  
```

Fig. 9 Sentence type identification

PSL Tree Generation

PSL tree generation is an essential step in the translation pipeline. In this step, the tree transformations are applied to the annotated tree generated in earlier stages using the defined PSL grammar. After the identification of sentence type, appropriate production from PSL grammar is chosen. As it is determined in the analysis part, there are mainly three operations that need to be performed on the tree nodes in order to convert the English tree to PSL tree. These operations are:

- Deletion of some nodes.
- Addition of new nodes.
- Alter node position in the tree.

Consider an example sentence “My dog is chasing a black cat” to understand the translation process of an English sentence to PSL sentence. An annotated tree and the final PSL tree have been presented in Figs. 10 and 11, respectively.

Different intermediate recursive steps of adding a node, deleting node, or altering node are applied on the source tree. The details of each step corresponding to the above example of present continuous tense are shown in Table 10 along with

appropriate rules of PSL grammar that have been applied, and also the functions applied on the source tree.

PSL Sentence Generation

The PSL tree shown in Fig. 11 is traversed using leave order traversal, and PSL sentence is generated as the outcome of the translation activity. The pseudo-code for this traversal is given in Fig. 12.

Translation System Prototype

In order to make this system useful for the deaf community, it is imperative to show the translated sentence in gesture-based representation. Before this research PSL, gestures have only been stored and exposed as one video or a GIF image for a word or phrase. To this end, the proposed system relies on the idea of a rich databank for sign language. Therefore, a sign databank has been created for more than 100 words that comprises of multiple representations of a word, including video gesture and GIF images. Thus, the translation system is integrated with the databank in a way that once the PSL sentence has been generated, the corresponding video or GIF image representation of each word is rendered from the databank in an appropriate order. Thus, the system shows the video gestures of all words in the PSL sentence in the correct order. This results into a real-time translation system that not only translates input natural language sentence into PSL but also presents the translated sentence as an avatar, as shown in Fig. 13. The purpose of showing the translated text in the form of avatar makes the translation model useable for the deaf community, and helps improving the cognitive abilities of the deaf subjects towards understanding the written text.

The integration of these two components builds a proof-of-concept system that implements a complete translation pipeline that translates input English language sentence into PSL gestures, thus aids the deaf community to improve their cognitive understanding in learning English language. As it is

only the proof of concept application, the proposed system only plays the gestures of words that are already present in the gesture data bank, while for the other words it uses finger spellings. However, the system can be seamlessly enriched by extending the databank with more GIF images representing more words.

Evaluation and Results

Due to the absence of conventions for writing PSL sentences in English, it is challenging to evaluate the accuracy of the proposed translation model for translating English sentences into equivalent PSL sentences. It is essential that the translation outcome of the proposed system must be acceptable as well as understandable to the deaf community. To assess the accuracy of our proposed grammar-based machine translation system, automatic and manual evaluations have been conducted.

Evaluation Mechanism

Automatic Evaluation In this kind of evaluation, the output generated by the translation system is compared with the gold standard data comprising of the correct translations of all sentences given as input to the translation system. The translation accuracy can be determined based on the similarity between the output generated by the system and the data stored against a particular sentence in the gold standard corpus [42]. As discussed earlier, PSL does not have any written standard available [2] so generating an evaluation corpus covering all categories of PSL sentences corresponding to their English equivalent version is a tough job which requires a lot of human effort and time. To the best of our knowledge, the corpus compiled in this research is the first sentence level corpus generated for PSL sentences corresponding to various linguistic categories of English sentences discussed in the data collection phase of this article.

Fig. 10 The annotated tree

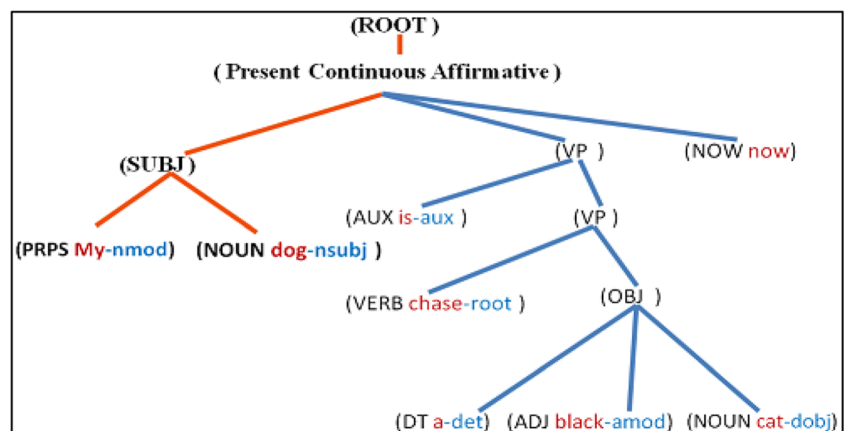
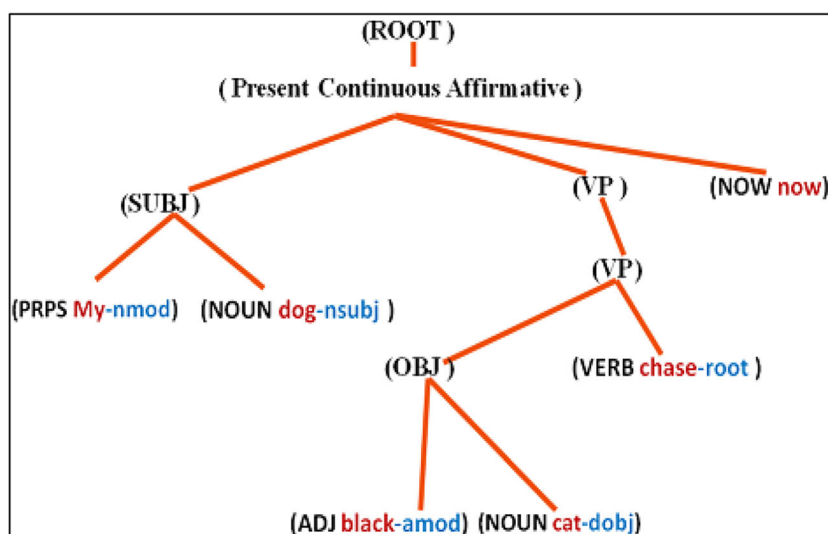


Fig. 11 Final PSL tree



Manual or User-Based Evaluation In such techniques, human experts are involved for evaluations. The accuracy of the translation output is evaluated based on the responses gathered by these experts who rated the output on different scales based on the syntactic and semantic understanding of the generated output produced by the automatic machine translation system.

Testing Corpus Generation According to English taxonomy, the sentences are categorised on the bases of tense, meaning, and structure. From contextual inquiry, it was observed that there are fourteen different types of sentences on the basis of tenses, three with respect to meanings, and four based on structural variants. The testing corpus developed in this research consists of a total of 2000 sentences covering all types of English sentences, as shown in Table 11.

Results of Experiments

Manual Evaluation In the first stage, experts assessed the system output manually to determine the system’s accuracy. It

was challenging to evaluate all 2000 sentences manually. Therefore, a subset comprising of nearly 500 sentences involving all types of sentences are manually assessed by two deaf persons and two bilingual experts of PSL and English. The evaluation results are divided into two groups valid and invalid. Sentences that preserve the syntactic and semantic structure of source language after translation are classified as valid sentences, whereas the ones which are not able to preserve the meaning of source language sentence were listed as invalid. Out of 500 sentences, 476 sentences listed as valid by both the evaluators, which makes the overall accuracy of the translation system about 95%. The 24 invalid sentences belong to complex and complex compound categories. The consistency of evaluations among different evaluators was very high with Kappa statistic of about 0.96.

Apart from different categories of sentences, we also tested the general applicability of our system on the story of a “thirsty crow”, which is a famous story in early school education. We enriched the databank to incorporate the GIF images for all the words included in this story. The deaf students were

Table 10 Derivation using PSL grammar

Sentence type	English grammar production	PSL production	Action
Present continuous affirmative	Present Continuous Affirmative	Present Continuous Affirmative	Add a child node in the tree having value “now”
	>SUBJ VP	>SUBJ VP	
	SUBJ	SUBJ	No Change
	>PRPS	>PRPS	
	NOUN	NOUN	
	VP → AUX	VP → VP	Remove Aux node from parent VP
	VP	VP → OBJ	
	VP → VERB OBJ	VERB	
	OBJ → DT ADJ NOUN	OBJ → ADJ NOUN	Remove DT child Node

```

1. Function: leaf_traversal ( TreeNode node)
2. Input:
3. TreeNode  $\leftarrow$  node
4. Begin:
5. String psl_sentence equals null
6. /* Call the recursive leaf_traversal function at root node*/
7. function leaf_traversal ( TreeNode node)
8.   if (node is a leaf node) then
9.     psl_sentece = psl_sentence + node.value
10.  else
11.    for  $\leftarrow$  1, to node.children
12.      leaf_traversal  $\leftarrow$  (Recursive Call)  $\leftarrow$  (ith node's child)
13.    end for
14.  end if
15. end function
16. End

```

Fig. 12 PSL sentence generation

able to understand the whole story, and they were elated to share that they had not experienced any such system before. Most of them stated that such system will help improving their cognitive abilities towards understanding written natural language text.

Results of Automatic Evaluation: Automatic evaluation is much less time consuming as compared with manual assessment. The latter evaluation technique is costly as it needs human subject involvement for verification. Another advantage of automatic evaluation is that it can be used to compare

the results of various other machine translation systems quite easily and without involving humans. In this work translation error rate (TER), word error rate (WER), and the bilingual evaluation understudy (BLEU) metrics are used to evaluate the performance of proposed machine translation system [43]. For this research work, the precision and BLEU score corresponding to reference, and candidate sentences are calculated using a 4-gram matching variant of BLEU [44]. The BLEU score ranges from 0 to 1. The exact match between reference and candidate sentence is represented by the value 1, while 0 shows no match. The word error rate is another performance measure used in machine translation systems. It is based on Levenshtein distance [4]. Once the translation is done, the TER is used to determine the extent of reordering of the candidate sentence in order to closely match the corresponding reference sentence [45].

The translated results of the system are evaluated based on reference sentences generated by bilingual human experts. The reference sentences cover all linguistic variants of English taxonomy discussed earlier in the data collection section. The result of 0.78, 0.10, and 0.15 was obtained using the BLEU, WER, and TER metrics, respectively. As during the grammar generation process, the variations in PSL sentence structures were restricted by choosing the most frequent word order as a standard rule; therefore, the rate of WER and TER is minimised, and similarly, the precision and the BLUE score is increased. Most of the incorrect sentences belonged to the complex and compound-complex categories.

Fig. 13 Avatar generation system

Enter English text here

You are going to school

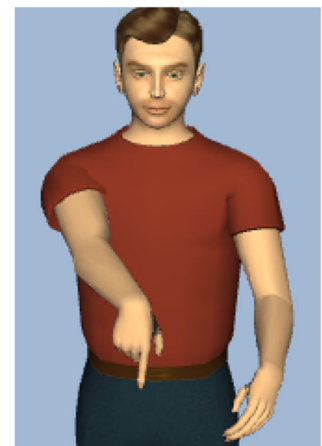
Translate & Play

Translate To PSL

you school go now

Debugger:

Playing sign :now
 Playing sign :go
 Playing sign :school
 Playing sign :you
 Play sequence you school go now.



now

Table 11 Testing corpus for evaluation

Type of sentence	Number of sentences
Simple present and past (positive, negative and interrogative)	250
Present indefinite, continuous, perfect, and perfect continuous (positive, negative, Aux and WH)	500
Past indefinite, continuous, perfect, and perfect continuous (positive, negative, Aux and WH)	500
Future indefinite, continuous, perfect, and perfect continuous (positive, negative, Aux and WH)	500
Simple, compound, complex, and compound complex	250
Total sentences	2000

Conclusion and Future Directions

This research presents a grammar-based machine translation model for translating English language sentences into equivalent PSL sentences. PSL is used by the deaf community of Pakistan, which has a different sentence structure, word order, and lexicon than English or Urdu. The contribution of this work is threefold. (1) A sentence level corpus is developed, which consists of 2000 sentences covering all English language sentence variations. Every English language sentence in this corpus is manually translated by involving native signers into PSL. This database has been made freely available for researchers. (2) The grammatical structure of PSL is formally transcribed. To this end, a systematic approach has been adopted involving data collection, analysis, and grammar rules generation, to define the grammar for PSL using CFG representation, which helps to textually represent PSL sentences. (3) A rule-based machine translation model is proposed to translate English text into PSL. The input English sentence to be translated into PSL is morphologically, syntactically, and semantically analysed. The outputs of the machine translation system result in sentences that satisfy the structure and grammar of PSL. Finally, the translation system has been evaluated by two domain experts and deaf subjects. The evaluation reveals that the system translates with a 0.78 BLEU score. A qualitative evaluation reveals that the developed translation system can help compensate for the cognitive hearing deficit of deaf people to understand English language.

In the future, we intend to improve the accuracy of the translation system by covering compound and compound-complex sentences which the current system is unable to translate correctly. Furthermore, such a system could generate more data to enable development of a natural to sign language translation system with increased generalisation capability, using state-of-the-art deep learning approaches (e.g. [46]). We also intend to design a parallel corpus for PSL gestures, since current corpora only contain words and their gestures. On the other hand, we aim to use signwriting notations along with videos, which are textual

representation of a gesture that can be read by machine and can further be converted into sign language gestures. This will help to extend our system so that it can convert the translated PSL sentence into an animated avatar that expresses word by word gestures of the translated sentence. Lastly, there is a need to extend the translation system to incorporate the pointing of pronouns in sign space [29, 30] by enriching sentence meta-information to ensure more accurate translation.

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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